EXPERIMENT 5: INFORMATION EXTRACTION USING SPACY

LEARNING OBJECTIVES:

- To understand information extraction from unstructured customer reviews.
- To preprocess and clean customer review data using Pandas.
- To perform tokenization, lemmatization, and Part-of-Speech (POS) tagging using spaCy.
- To identify important linguistic categories such as nouns, verbs, adjectives, and adverbs.
- To perform Named Entity Recognition (NER) on customer reviews.
- To extract entities such as PERSON, ORG, DATE, MONEY, QUANTITY, and PRODUCT.
- To organize extracted information into structured DataFrames.
- To develop a reusable function for extracting linguistic and named-entity information from new reviews.

METHODS / PYTHON LIBRARIES USED:

Library/method	Purpose	Reason for use
spaCy	Natural Language Processing	Provides an efficient NLP pipeline for text analysis
en_core_web_sm	English NLP Model	Performs tokenization, POS tagging and NER
Tokenization	Splits text into individual words/tokens	Helps analyze the components of customer reviews
Lemmatization	Converts words into their base form	Helps normalize words for analysis
Counter	Counts POS category frequencies	Helps determine the frequency of major POS categories

PROGRAM CODE:

```
import pandas as pd
import spacy
from collections import Counter

nlp = spacy.load("en_core_web_sm")

df = pd.read_csv("Reviews.csv", engine="python", on_bad_lines="skip")
df = df[["Text"]].dropna().drop_duplicates().head(10000)

print("Dataset Shape:", df.shape)

for review in df["Text"].head(5):
    doc = nlp(review)
    print("\nReview:", review)
    print("Tokens:", [t.text for t in doc])
    print("Lemmas:", [t.lemma_ for t in doc])
    print("POS:", [(t.text, t.pos_) for t in doc])

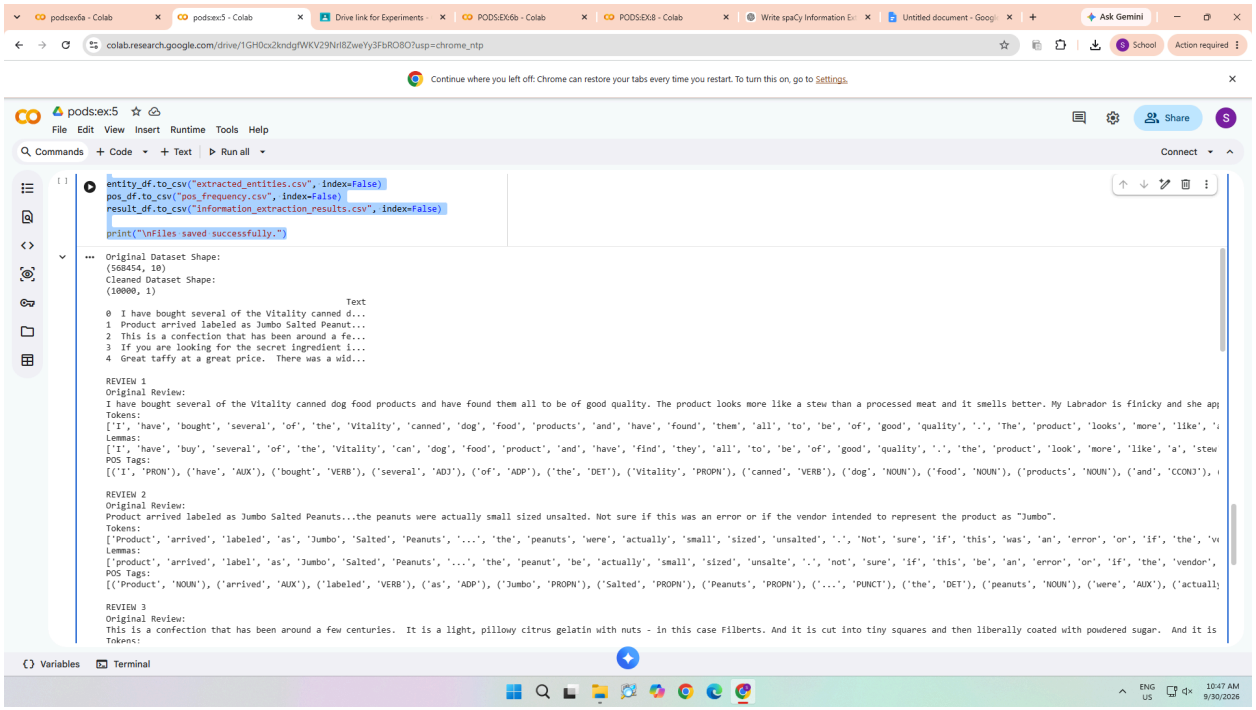
counter = Counter()

for review in df["Text"]:
    doc = nlp(review)
    for t in doc:
        if t.pos_ in ["NOUN", "VERB", "ADJ", "ADV"]:
            counter[t.pos_] += 1

reviews = [
    "I bought this product from Amazon last week.",
    "The Apple coffee maker costs $50.",
    "John ordered 2 boxes of snacks from Walmart."
]

for review in reviews:
    print("\nReview:", review)
    print(extract_information(review))
```

OUTPUT:



```
entity_df.to_csv("extracted_entities.csv", index=False)
pos_df.to_csv("pos_frequency.csv", index=False)
result_df.to_csv("information_extraction_results.csv", index=False)

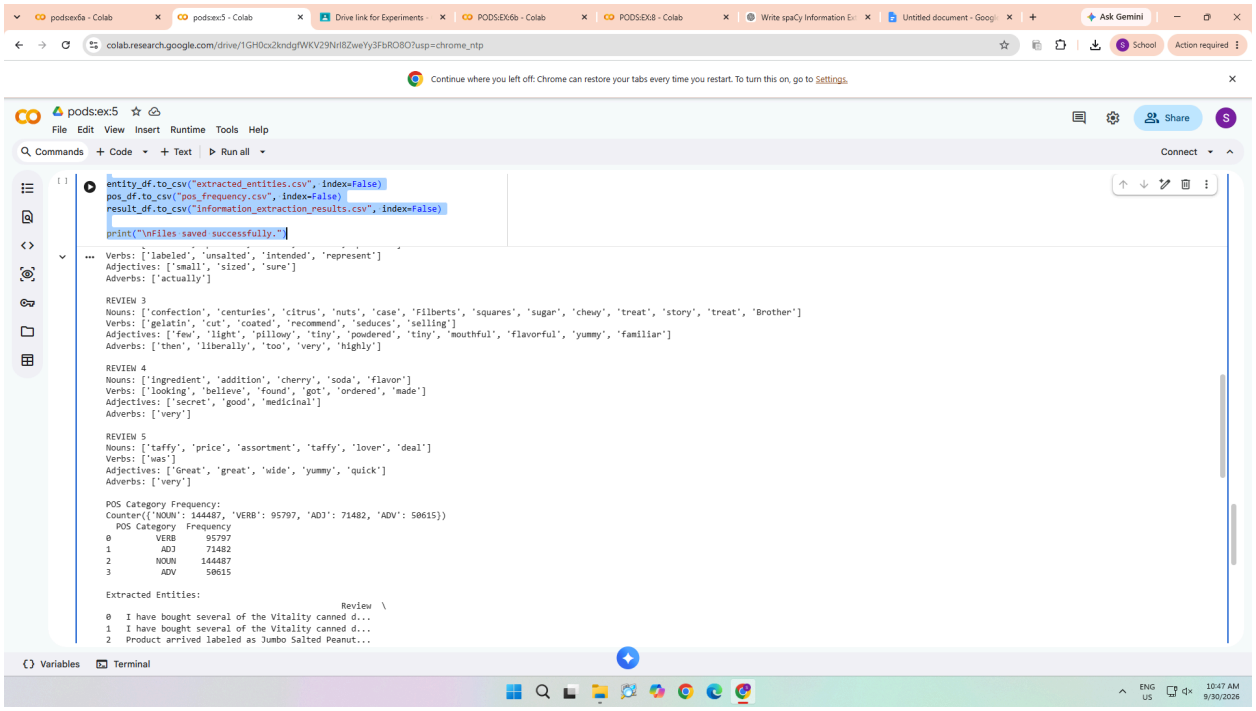
print("\nFiles saved successfully.")
```

```
Original Dataset Shape:
(568454, 10)
Cleaned Dataset Shape:
(10000, 1)
                                                Text
0  I have bought several of the Vitality canned d...
1  Product arrived labeled as Jumbo Salted Peanut...
2  This is a confection that has been around a fe...
3  If you are looking for the secret ingredient i...
4  Great taffy at a great price.  There was a wid...

REVIEW 1
Original Review:
I have bought several of the Vitality canned dog food products and have found them all to be of good quality. The product looks more like a stew than a processed meat and it smells better. My Labrador is finicky and she ap
Tokens:
['I', 'have', 'bought', 'several', 'of', 'the', 'Vitality', 'canned', 'dog', 'food', 'products', 'and', 'have', 'found', 'them', 'all', 'to', 'be', 'of', 'good', 'quality', '.', 'The', 'product', 'looks', 'more', 'like', '
Lemmas:
['I', 'have', 'buy', 'several', 'of', 'the', 'Vitality', 'can', 'dog', 'food', 'product', 'and', 'have', 'find', 'they', 'all', 'to', 'be', 'of', 'good', 'quality', '.', 'the', 'product', 'look', 'more', 'like', 'a', 'stew
POS Tags:
[('I', 'PRON'), ('have', 'AUX'), ('bought', 'VERB'), ('several', 'ADJ'), ('of', 'ADP'), ('the', 'DET'), ('Vitality', 'PROPN'), ('canned', 'VERB'), ('dog', 'NOUN'), ('food', 'NOUN'), ('products', 'NOUN'), ('and', 'CCONJ'),

REVIEW 2
Original Review:
Product arrived labeled as Jumbo Salted Peanuts...the peanuts were actually small sized unsalted. Not sure if this was an error or if the vendor intended to represent the product as "Jumbo".
Tokens:
['Product', 'arrived', 'labeled', 'as', 'Jumbo', 'Salted', 'Peanuts', '...', 'the', 'peanuts', 'were', 'actually', 'small', 'sized', 'unsalted', '.', 'Not', 'sure', 'if', 'this', 'was', 'an', 'error', 'or', 'if', 'the', 'v
Lemmas:
['product', 'arrived', 'label', 'as', 'Jumbo', 'Salted', 'Peanuts', '...', 'the', 'peanut', 'be', 'actually', 'small', 'sized', 'unsalte', '.', 'not', 'sure', 'if', 'this', 'be', 'an', 'error', 'or', 'if', 'the', 'vendor',
POS Tags:
[('Product', 'NOUN'), ('arrived', 'AUX'), ('labeled', 'VERB'), ('as', 'ADP'), ('Jumbo', 'PROPN'), ('Salted', 'PROPN'), ('Peanuts', 'PROPN'), ('...', 'PUNCT'), ('the', 'DET'), ('peanuts', 'NOUN'), ('were', 'AUX'), ('actuall

REVIEW 3
Original Review:
This is a confection that has been around a few centuries.  It is a light, pillowy citrus gelatin with nuts - in this case Filberts. And it is cut into tiny squares and then liberally coated with powdered sugar.  And it is
Tokens:
```



```
entity_df.to_csv("extracted_entities.csv", index=False)
pos_df.to_csv("pos_frequency.csv", index=False)
result_df.to_csv("information_extraction_results.csv", index=False)

print("\nFiles saved successfully.")
```

```
Verbs: ['labeled', 'unsalted', 'intended', 'represent']
Adjectives: ['small', 'sized', 'sure']
Adverbs: ['actually']

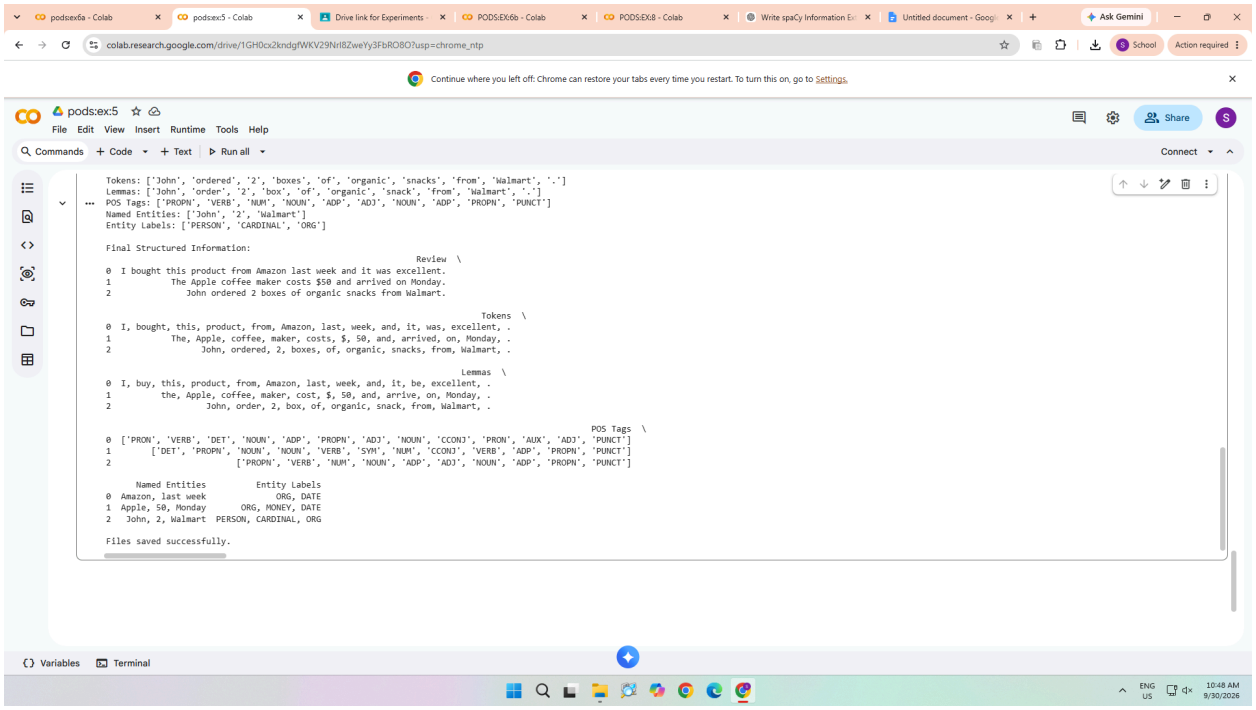
REVIEW 3
Nouns: ['confection', 'centuries', 'citrus', 'nuts', 'case', 'Filberts', 'squares', 'sugar', 'chewy', 'treat', 'story', 'treat', 'Brother']
Verbs: ['gelatin', 'cut', 'coated', 'recommend', 'seduces', 'selling']
Adjectives: ['few', 'light', 'pillowy', 'tiny', 'powdered', 'tiny', 'mouthful', 'flavorful', 'yummy', 'familiar']
Adverbs: ['then', 'liberally', 'too', 'very', 'highly']

REVIEW 4
Nouns: ['ingredient', 'addition', 'cherry', 'soda', 'flavor']
Verbs: ['looking', 'believe', 'found', 'got', 'ordered', 'made']
Adjectives: ['secret', 'good', 'medicinal']
Adverbs: ['very']

REVIEW 5
Nouns: ['taffy', 'price', 'assortment', 'taffy', 'lover', 'deal']
Verbs: ['was']
Adjectives: ['Great', 'great', 'wide', 'yummy', 'quick']
Adverbs: ['very']

POS Category Frequency:
Counter({'NOUN': 144487, 'VERB': 95797, 'ADJ': 71482, 'ADV': 50615})
  POS Category  Frequency
0         VERB      95797
1          ADJ      71482
2         NOUN     144487
3          ADV      50615

Extracted Entities:
                                              Review  \
0  I have bought several of the Vitality canned d...
1  I have bought several of the Vitality canned d...
2  Product arrived labeled as Jumbo Salted Peanut...
```



LEARNING OUTCOME:

- Load and clean customer review datasets.
- Apply spaCy NLP pipelines to unstructured text.
- Perform tokenization and lemmatization.
- Identify nouns, verbs, adjectives, and adverbs using POS tagging.
- Calculate the frequency of major POS categories.
- Apply Named Entity Recognition (NER) to customer reviews.
- Extract entities such as persons, organizations, dates, money, and quantities.
- Create reusable NLP functions for information extraction.
- Convert unstructured customer reviews into structured information for customer feedback analysis.